Data Extraction for Machine Learning: Methods, Sources and Pitfalls

Programming Project P1 — DataHunt (Track: *GovAI*)

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September 26, 2026

Abstract

One paragraph: the problem, the sources used (including Eurostat and one unstructured source), the pipeline, the baseline result, and the main lesson about data extraction.

1 Introduction

Data acquisition is the first stage of every ML project [6]; errors made here propagate downstream as "data cascades" [7].

2 Background and Related Work

Survey extraction methods. Compare them in Table 1.

Table 1: Data extraction methods compared.

Method	Stability	Effort	Legal risk	Typical format
File download	high	low	low	CSV, Parquet
REST / SDMX API	high	low–medium	low	JSON, SDMX, JSON-stat
Web scraping	low	medium–high	medium–high	HTML
Document parsing	medium	medium	depends	PDF, DOCX
Media collection	medium	high	high (copyright, GDPR)	JPG, WAV

3 Legal and Ethical Framework

Licences, GDPR, the Open Data Directive [1], the AI Act [2].

4 Case Study: Our Pipeline

Architecture figure, sources table, provenance design following FAIR [9], tidy data [8], Eurostat API [3].

5 From Data to a Baseline Model

6 Data Quality and Threats to Validity

Discuss dataset documentation [4] and dataset practice [5].

7 Conclusions

A Datasheet

B AI-use Statement

State which AI tools were used, for what, and how every generated claim and reference was verified.

References

[1] European Parliament and Council of the European Union. Directive (EU) 2019/1024 on open data and the re-use of public sector information. Official Journal of the European Union, L 172, 2019. URL https://eur-lex.europa.eu/eli/dir/2019/1024/oj.

[2] European Parliament and Council of the European Union. Regulation (EU) 2024/1689 laying down harmonised rules on artificial intelligence (AI act), 2024. URL https://eur-lex.europa.eu/eli/reg/2024/1689/oj.

[3] Eurostat. Eurostat API – getting started. Eurostat online help, 2026. URL https://wikis.ec.europa.eu/display/EUROSTATHELP/API+-+Getting+started.

[4] Timnit Gebru, Jamie Morgenstern, Briana Vecchione, Jennifer Wortman Vaughan, Hanna Wallach, Hal Daumé III, and Kate Crawford. Datasheets for datasets. *Communications of the ACM*, 64(12):86–92, 2021. doi: 10.1145/3458723.

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[6] Yuji Roh, Geon Heo, and Steven Euijong Whang. A survey on data collection for machine learning: A big data–AI integration perspective. *IEEE Transactions on Knowledge and Data Engineering*, 33(4):1328–1347, 2021. doi: 10.1109/TKDE.2019.2946162.

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[8] Hadley Wickham. Tidy data. *Journal of Statistical Software*, 59(10), 2014. doi: 10.18637/jss.v059.i10.

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